Mole Concept

BITSAT · Nonofficial practice
Attempt: dead6e7b-b490-49ea-a6e6-07823f5b7da6
Started: 2026-10-17T14:27:05.192567+00:00
Content version: 1 · Key version: 1
Score version: original

Score: -1 / 150 · Percentage: -0.67% (not a percentile)
Correct: 0 · Incorrect: 1 · Skipped: 49
Dropped: 0 · Bonus-to-all: 0
Attempt rate: 2.0% · Accuracy: 0.0%
Accuracy = correct / attempted; dropped and bonus items excluded.
Penalty marks: 1 · Elapsed: 4 seconds
Rules: +3 correct, -1 wrong, 0 unanswered
Numerical: Single-correct MCQ: select A, B, C or D. No numeric-entry questions.

Complete record: all 50 questions in original delivery order.
Skipped questions and supplied answer keys are included. No worked solutions were supplied.
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Frozen profile: 01bdc01e85862d9104bcbd315feec72325658ba261ef280493fa656a4d25221a

Question 1 · Chemistry

Your response: A | Accepted answer(s): C
Marks: -1 | Incorrect | Reason: incorrect_response

Question

01. Two liquids have densities 1.00 and 1.50 $\mathrm{g\,mL^{-1}}$. They mix without volume change to density 1.20 $\mathrm{g\,mL^{-1}}$. What fraction of the mixture mass came from the denser liquid?

(A) 0.6 (B) 0.75

(C) 0.5 (D) 0.4

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 5

Supplied answer key (no worked solution)

01	**C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 2 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

02. A mixture contains only CO and CO_2. Its mean molar mass is $36\,g\,mol^{-1}$. How many moles of oxygen atoms are present in 18 g of the mixture?

(A) 1.5
(B) 1
(C) 0.5
(D) 0.75

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 5

Supplied answer key (no worked solution)

02 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 3 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

03. A compound contains only C, H, O. Complete combustion of 0.60 g gives 0.88 g CO_2 and 0.36 g water. If its molar mass is 90 g mol^{-1}, the number of hydrogen atoms per molecule is

(A) 4 (B) 6

(C) 2 (D) 12

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 5

Supplied answer key (no worked solution)

03 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 4 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

04. A 10.0 g sample contains $CaCO_3$ and inert material. It is treated with 0.200 mol HCl; the unreacted acid needs 0.0800 mol NaOH. Percentage by mass of $CaCO_3$ is (molar mass 100)

(A) 60
(B) 80
(C) 100
(D) 40

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 5

Supplied answer key (no worked solution)

04	**A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 5 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

05. A hydrate $MSO_4 \cdot xH_2O$ has 40.0% water by mass. Its anhydrous salt has molar mass $162\,g\,mol^{-1}$. The number of oxygen atoms per formula unit of the hydrate is

(A) 10 (B) 8

(C) 4 (D) 6

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 5

Supplied answer key (no worked solution)

05 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 6 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

06. A gaseous hydrocarbon consumes five times its own volume of oxygen on complete combustion and forms three times its volume of CO_2. All gas volumes are at the same T, P. Hydrogen atoms per molecule are

(A) 8 (B) 4

(C) 6 (D) 10

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 6

Supplied answer key (no worked solution)

06 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 7 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

07. 2.00 g of a mixture of NaOH and Na_2CO_3 needs 0.04509434 mol HCl for complete neutralization. Mass of NaOH in g is (molar masses 40, 106)

(A) 1.60 (B) 0.40

(C) 0.80 (D) 1.20

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 6

Supplied answer key (no worked solution)

07 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 8 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

08. A reaction $2A + 3B \rightarrow 4C$ is run with 0.60 mol A and 0.60 mol B. The isolated C contains 0.48 mol. What is the percentage yield?

(A) 60 (B) 75

(C) 40 (D) 80

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 6

Supplied answer key (no worked solution)

08 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 9 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

09. A 20.0 g mixture of $CaCO_3$ and an inert solid loses 4.40 g on complete heating. What percentage of the final solid mass is CaO? ($CaCO_3$ =100, CaO =56)

(A) 56

(B) 28

(C) 35.897436

(D) 64.102564

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 6

Supplied answer key (no worked solution)

09 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 10 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

10. A mixture contains equal numbers of CH_4 and C_2H_6 molecules. What is the mass percentage of carbon in the mixture? C=12, H=1.

(A) 78.26087
(B) 80
(C) 75
(D) 85.714286

11–40 / New middle-to-hard practice

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 6

Supplied answer key (no worked solution)

10 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 11 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

11. A mixture contains only Na_2CO_3 and $NaHCO_3$. One aliquot consumes 25.0 mL of 0.100 M HCl to the phenolphthalein end point and 65.0 mL in total to the methyl-orange end point. The mole fraction of Na_2CO_3 in the original mixture is

(A) $\frac{3}{8}$ (B) $\frac{10}{13}$

(C) $\frac{5}{13}$ (D) $\frac{5}{8}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 6

Supplied answer key (no worked solution)

11 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 12 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

12. An 8.20 g mixture of MgO and $MgCO_3$ consumes 0.300 mol HCl completely. On strong heating a fresh identical sample to constant mass, the mass lost is (Mg=24, C=12, O=16)

(A) 4.40 g
(B) 2.20 g
(C) 1.10 g
(D) 3.30 g

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 6

Supplied answer key (no worked solution)

12 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 13 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

13. A pure hydrate $CuSO_4 \cdot xH_2O$ weighs 25.00 g. Complete dehydration leaves 16.00 g, and all sulfate in that residue gives 23.30 g $BaSO_4$. With $M(BaSO_4) = 233$, the number of water molecules per copper ion is

(A) 10 (B) 4

(C) 6 (D) 5

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 6

Supplied answer key (no worked solution)

13 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 14 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

14. A 0.890 g compound containing C, H, N and O gives 1.320 g CO_2 and 0.630 g water on complete combustion. A separate identical sample gives 112 mL N_2 at STP. Its molar mass is 89. The molecular formula is (STP molar volume 22.4 L)

(A) $C_3H_9N_2O$ (B) $C_3H_7NO_2$

(C) $C_2H_5NO_2$ (D) C_3H_7NO

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 7

Supplied answer key (no worked solution)

14 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 15 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

15. A 1.00 L mixture of ethene and ethyne is hydrogenated completely using 1.60 L H_2. All volumes refer to the same temperature and pressure. The original gas mixture is then considered before hydrogenation: the oxygen volume needed for its complete combustion is

(A) 2.40 L
(B) 2.60 L
(C) 3.00 L
(D) 2.70 L

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 7

Supplied answer key (no worked solution)

15 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 16 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

16. A gaseous hydrocarbon has density 1.25 $\mathrm{g\,L^{-1}}$ at STP. A 0.10 mol sample consumes 0.30 mol O_2 completely. The number of hydrogen atoms in one molecule is (STP molar volume 22.4 L)

(A) 2 (B) 4

(C) 6 (D) 8

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 7

Supplied answer key (no worked solution)

16 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 17 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

17. A 9.88 g mixture of $CaCO_3$ and $MgCO_3$ is heated completely. It loses 4.84 g. The mole fraction of $CaCO_3$ in the original carbonate mixture is (molar masses 100 and 84)

(A) $\frac{4}{11}$ (B) $\frac{2}{5}$

(C) $\frac{7}{11}$ (D) $\frac{3}{5}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 7

Supplied answer key (no worked solution)

17 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 18 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

18. A mixture of FeO and Fe_2O_3 contains 0.150 mol Fe atoms. Complete reduction gives 8.40 g Fe and removes 2.80 g oxygen. The mole fraction of FeO among the two oxides is (Fe=56, O=16)

(A) $\frac{4}{5}$ (B) $\frac{1}{5}$
(C) $\frac{3}{5}$ (D) $\frac{2}{3}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 7

Supplied answer key (no worked solution)

18 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 19 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

19. One mole of an element contains isotopes of mass numbers 20, 21 and 22. Their number ratio is $x : 1 : 2$, and the mean mass number is 21.0. The fraction of total mass contributed by isotope 22 is

(A) $\frac{22}{105}$
(B) $\frac{44}{105}$
(C) $\frac{11}{25}$
(D) $\frac{2}{5}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 7

Supplied answer key (no worked solution)

19 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 20 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

20. A_2B_3 contains 40% A by mass. AB_2 containing the same isotopes has molar mass 180. The atomic molar mass of B is

(A) 90 (B) 45

(C) 120 (D) 60

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 7

Supplied answer key (no worked solution)

20 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 21 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

21. A 0.500 mol feed of A and 0.600 mol B undergoes $2A + 3B \rightarrow 4C$. All limiting reagent reacts, but only 80.0% of the C formed is recovered. The recovered C is used in $C + 2D \rightarrow E$ with 0.500 mol D. Maximum E obtained is

(A) 0.640 mol
(B) 0.500 mol
(C) 0.250 mol
(D) 0.320 mol

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 7

Supplied answer key (no worked solution)

21	**C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 22 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

22. A solution is 20.0% by mass in H_2SO_4 and has density 1.225 g mL^{-1}. A 20.0 mL aliquot is diluted and fully neutralised. Moles of NaOH needed are ($M(H_2SO_4) = 98$)

(A) $\frac{1}{20}$ (B) $\frac{1}{40}$
(C) $\frac{1}{10}$ (D) $\frac{1}{5}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 8

Supplied answer key (no worked solution)

22 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 23 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

23. A 9.20 g alloy contains zinc and aluminium only. It gives 5.60 L hydrogen at STP with excess dilute HCl. A fresh identical sample is completely converted to ZnO and Al_2O_3 without metal loss. The ZnO mass fraction in the oxide mixture is (Zn=65, Al=27, O=16)

(A) $\frac{27}{40}$ (B) $\frac{17}{44}$
(C) $\frac{27}{44}$ (D) $\frac{13}{22}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 8

Supplied answer key (no worked solution)

23 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 24 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

24. A 24.0 g mixture contains only methane and oxygen. On complete ignition followed by cooling, exactly 9.00 g liquid water is collected and oxygen remains. The mass of unreacted oxygen is

(A) 4.00 g (B) 2.00 g

(C) 0 (D) 8.00 g

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 8

Supplied answer key (no worked solution)

24 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 25 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

25. A solid mixture contains NaCl and KCl only. A 6.650 g sample gives 14.35 g AgCl with excess $AgNO_3$. The mole fraction of KCl is (Na=23, K=39, Cl=35.5, Ag=108)

(A) $\frac{3}{4}$ (B) $\frac{1}{4}$
(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 8

Supplied answer key (no worked solution)

25 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 26 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

26. A 1.000 g sample contains anhydrous Na_2CO_3 and water only. It consumes 15.0 mL of 1.00 M HCl for complete neutralisation. The sample is dried without decomposing carbonate. Percentage mass loss is $(M(Na_2CO_3) = 106)$

(A) 20.5%
(B) 79.5%
(C) 10.25%
(D) 41.0%

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 8

Supplied answer key (no worked solution)

26	**A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 27 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

27. A solution initially contains 0.100 mol $AgNO_3$. It is treated with 0.0400 mol $CaCl_2$, and the precipitate is removed completely. The filtrate is then treated with excess NaBr. Mass of the second precipitate is (Ag=108, Br=80)

(A) 18.80 g
(B) 7.52 g
(C) 3.76 g
(D) 15.04 g

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 8

Supplied answer key (no worked solution)

27	**C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 28 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

28. A 0.200 mol mixture of Na_2SO_4 and Na_3PO_4 contains 0.500 mol Na atoms. The mixture is treated with excess $BaCl_2$. Total moles of barium ions incorporated in both precipitates are

(A) $\frac{1}{4}$ (B) $\frac{1}{5}$

(C) $\frac{1}{2}$ (D) $\frac{3}{10}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 8

Supplied answer key (no worked solution)

28 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 29 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

29. A hydrocarbon mixture contains 0.200 mol methane and an unknown amount of ethane. Complete combustion yields 0.800 mol CO_2. The amount of water formed is

(A) 1.00 mol
(B) 1.60 mol
(C) 1.30 mol
(D) 0.800 mol

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 8

Supplied answer key (no worked solution)

29 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 30 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

30. A 2.00 mol mixture of N_2 and H_2 has mean molar mass 8.00. It is passed over an ammonia catalyst with complete conversion of the limiting reactant, then cooled to remove all ammonia. Moles of gas left are (N=14, H=1)

(A) $\frac{8}{13}$
(B) $\frac{2}{13}$
(C) $\frac{6}{13}$
(D) $\frac{4}{13}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 9

Supplied answer key (no worked solution)

30 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 31 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

31. A salt M_2X_3 contains metal M in the $+3$ state and X in the -2 state. Its metal mass fraction is 0.400. If the equivalent mass of M is 18.0, the atomic molar mass of X is

(A) 36 (B) 54

(C) 27 (D) 81

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 9

Supplied answer key (no worked solution)

31 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 32 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

32. A gaseous mixture of CO and H_2 contains equal masses of the two gases. Its 15.0 mol sample is converted completely by $CO + 2H_2 \rightarrow CH_3OH$. Moles of hydrogen remaining are

(A) 10 (B) 11

(C) 13 (D) 12

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 9

Supplied answer key (no worked solution)

32 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 33 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

33. 0.100 mol $Ca_3(PO_4)_2$ is reacted with excess H_2SO_4 to form only H_3PO_4 and $CaSO_4$. The phosphoric acid is then completely neutralised with NaOH. Total moles of NaOH required are

(A) $\frac{3}{5}$ (B) $\frac{1}{5}$

(C) $\frac{3}{10}$ (D) $\frac{9}{10}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 9

Supplied answer key (no worked solution)

33 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 34 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

34. A 27.8 g sample of $FeSO_4 \cdot xH_2O$ leaves 15.2 g anhydrous $FeSO_4$ on heating without redox change. The fraction of all original oxygen atoms that remain in the solid is $(M(FeSO_4) = 152)$

(A) $\frac{7}{11}$ (B) $\frac{4}{11}$
(C) $\frac{4}{7}$ (D) $\frac{8}{11}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 9

Supplied answer key (no worked solution)

34 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 35 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

35. In an oxygen sample, 80.0% of atoms are ^{16}O and the rest are ^{18}O. Atoms pair randomly into molecules. The mass fraction of $^{16}O^{18}O$ molecules in the gas is

(A) $\frac{17}{50}$ (B) $\frac{68}{205}$
(C) $\frac{8}{25}$ (D) $\frac{4}{25}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 9

Supplied answer key (no worked solution)

35 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 36 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

36. A salt contains only Fe^{2+}, Fe^{3+} and oxide ions. A sample contains 0.300 mol Fe and 0.400 mol O. The fraction of iron atoms in the +3 state is

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$
(C) $\frac{3}{4}$ (D) $\frac{1}{3}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 9

Supplied answer key (no worked solution)

36 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 37 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

37. A gas mixture contains 0.100 mol N_2O_4 and 0.100 mol NO_2. Exactly half the initial N_2O_4 dissociates into NO_2. The ratio of final gas-molecule count to final nitrogen-atom count is

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$
(C) $\frac{3}{5}$ (D) $\frac{5}{6}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 9

Supplied answer key (no worked solution)

37 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 38 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

38. An alloy contains copper and zinc only. A 12.8 g sample evolves 2.24 L H_2 at STP with excess dilute HCl; copper does not react. The residual copper is oxidised to CuO. Mass of CuO formed is (Cu=63, Zn=65, O=16)

(A) 15.80 g
(B) 6.30 g
(C) 7.90 g
(D) 8.10 g

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 10

Supplied answer key (no worked solution)

38 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 39 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

39. A mixture contains $Fe_2(SO_4)_3$ and $MgSO_4$ only, with 0.500 mol sulfate ions and 0.400 mol metal ions. Both metals are precipitated completely as hydroxides. The mass fraction of $Fe(OH)_3$ in the precipitate is (Fe=56, Mg=24, O=16, H=1)

(A) $\frac{107}{223}$
(B) $\frac{1}{2}$
(C) $\frac{58}{165}$
(D) $\frac{107}{165}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 10

Supplied answer key (no worked solution)

39 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 40 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

40. A 5.60 g mixture of CO and CO_2 occupies 3.36 L at STP. It is passed through hot carbon so that all CO_2 becomes CO by $CO_2 + C \rightarrow 2CO$. The mass of carbon consumed is

(A) 1.20 g
(B) 0.900 g
(C) 1.05 g
(D) 0.600 g

41–50 / Preserved hard practice

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 10

Supplied answer key (no worked solution)

40 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 41 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

41. A primary acyclic saturated monoamine produces 0.0100 mol N_2 in Dumas analysis from a 1.180 g sample. Assuming quantitative nitrogen recovery and formula $C_nH_{2n+3}N$, the number of constitutional primary amines with this formula is

(A) 3 (B) 4

(C) 2 (D) 1

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Supplied answer key (no worked solution)

41 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 42 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

42. A mixture contains ethanol and propan-1-ol only. Its mass is 5.80 g and excess sodium liberates 0.0500 mol H_2. Complete combustion of the original mixture gives CO_2 in amount (mol)

(A) 0.28571429
(B) 0.25
(C) 0.3
(D) 0.2

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 10

Supplied answer key (no worked solution)

42 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 43 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

43. A 0.100 mol mixture of propane and ethene consumes 0.380 mol oxygen on complete combustion. When a fresh equal sample is hydrogenated quantitatively, moles of H_2 required are

(A) 0.06
(B) 0.08
(C) 0.02
(D) 0.04

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 10

Supplied answer key (no worked solution)

43 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 44 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

44. A 0.0200 mol sample of a chromium complex has empirical composition $CrCl_3 \cdot 6H_2O$. Excess $AgNO_3$ immediately precipitates 0.0400 mol AgCl. The complex is octahedral with only water and chloride ligands. How many moles of coordinated water are in the sample?

(A) 0.06
(B) 0.1
(C) 0.08
(D) 0.12

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 10

Supplied answer key (no worked solution)

44 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 45 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

45. A 2.060 g sample containing NaOH, Na_2CO_3 and an inert solid is titrated with HCl. The phenolphthalein endpoint needs 0.0300 mol HCl; complete neutralization needs 0.0400 mol. The mass percentage of Na_2CO_3 is (NaOH =40, Na_2CO_3 =106)

(A) 25.728155
(B) 77.184466
(C) 51.456311
(D) 48.543689

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 11

Supplied answer key (no worked solution)

45 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 46 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

46. A 4.22 g mixture of $CaCO_3$ and $MgCO_3$ is fully decomposed. It releases 0.0450 mol CO_2. The final oxide residue is completely neutralized by HCl. What mass of MgO was present in that residue? Molar masses: $CaCO_3$ =100, $MgCO_3$ =84, MgO =40.

(A) 0.4
(B) 0.7
(C) 1.8
(D) 1.1

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Supplied answer key (no worked solution)

46 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 47 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

47. A mixture of Fe^{2+} and Fe^{3+} contains 0.0200 mol total iron. Oxidizing all Fe^{2+} to Fe^{3+} needs 0.00300 mol dichromate in acid. Reducing all iron in a fresh identical sample to metal requires moles of electrons equal to

(A) 0.04
(B) 0.042
(C) 0.036
(D) 0.06

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 11

Supplied answer key (no worked solution)

47 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 48 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

48. A liquid containing C, H, O only gives C: H atom ratio 1: 2 by combustion. Its oxygen mass fraction is 8/29. Its vapour molar mass is $58\,\mathrm{g\,mol^{-1}}$. It gives a silver mirror but no iodoform precipitate. Number of carbon atoms in its molecule is

(A) 2 (B) 6

(C) 4 (D) 3

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 11

Supplied answer key (no worked solution)

48 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 49 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

49. A 0.100 mol mixture of $FeSO_4$ and $Fe_2(SO_4)_3$ is treated with excess $BaCl_2$ to precipitate sulfate. A separate identical sample consumes 0.0100 mol acidic dichromate to oxidize Fe^{2+}. Moles of $BaSO_4$ formed are

(A) 0.12 (B) 0.1

(C) 0.24 (D) 0.18

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 11

Supplied answer key (no worked solution)

49 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12

Question 50 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

50. A 0.100 mol mixture of an alkane C_nH_{2n+2} and the corresponding alkene C_nH_{2n} has mass 5.70 g and consumes 0.0500 mol H_2 on complete hydrogenation. Total oxygen required for complete combustion of the original sample is (mol)

(A) 0.6
(B) 0.65
(C) 0.675
(D) 0.625

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Supplied answer key (no worked solution)

50 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 12